



LONI IMAGE & DATA ARCHIVE USER MANUAL

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1. INTRODUCTION

The Image and Data Archive (IDA) is a secure online resource for archiving, exploring and sharing neuroscience data including MRI, PET, SPECT, EEG and other modalities. A flexible data de-identification engine and encrypted file transmission help ensure compliance with patient-privacy regulations. All data are stored on redundant servers with daily and weekly on- and off-site backups. Archiving data in the IDA is simple and secure. As part of the archiving process relevant metadata are extracted from files, allowing data to be searched within moments of being archived. The archive protects data from unauthorized access and allows investigators to share data with other researchers.

2. ABBREVIATIONS

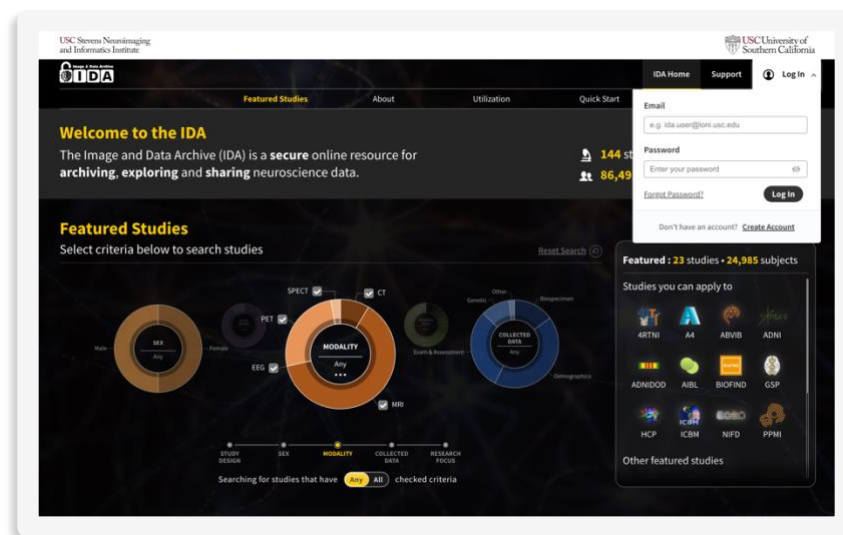
CT	Computed Tomography
DICOM	Digital Imaging and Communication in Medicine
DTI	Diffusion Tension Imaging
EEG	Electroencephalogram
FMRI	Functional Magnetic Resonance Imaging
MRA	Magnetic Resonance Angiography
MRI	Magnetic Resonance Imaging
PET	Positron Emission Tomography
SPECT	Single Photon Emission Computed Tomography

3. USER REGISTRATION

Accessing data in the IDA requires a user account. Please follow the steps below to register for an IDA account. After registration, you will immediately have access to all publicly available studies located in the SELECT STUDY menu. However, many studies require a separate data use application. Please see the ‘Studies you can apply to’ section for more information on requesting access to data from other studies.

Steps to Register for an IDA account:

1. At the top right corner of the [IDA home page](#), click Log In, then click **Create Account**.
2. Complete the new account registration process (3 steps).



- a. Enter your email address and click **Continue**. A security code will be emailed to you from ida@loni.usc.edu (you may need to add this email to your safe sender list).

Create IDA Account

1 Enter Email — 2 Verify Email — 3 Create Account

Enter Your Email

Please enter your email address below and you will receive a security code to verify your email address.

Email

Retype Email

☐ I'm not a robot

 reCAPTCHA Privacy - Terms

Continue

b. Enter the security code emailed to you and click **Continue**.

Create IDA Account

✓ Enter Email — 2 Verify Email — 3 Create Account

Verify Your Email

Please enter the security code that we sent to @usc.edu

Security Code

[Resend Code \(118\)](#)

i If you haven't received a security code from **ida@loni.usc.edu**, please check your SPAM folder. We recommend adding ida@loni.usc.edu to your safe sender list to ensure you receive the security code.

Note: The security code will expire once it is used or after your web browser is closed. If you receive more than one security code, please use the latest one.

Back **Continue**

- c. Complete the Create IDA Account form and click **Complete**.

The screenshot shows the 'Create IDA Account' form at the 'Create Account' step. The progress bar at the top indicates 'Enter Email' and 'Verify Email' are complete, while 'Create Account' is the current step. The form is titled 'Your Information' and includes a sub-header 'Section 1 - Create Password' with a note that passwords must be at least 16 characters. It features two password input fields: 'New Password' and 'Retype Password', both with a strength indicator (N/A). Below this is 'Section 2 - Your information' with fields for 'First Name*', 'Last Name*', 'Institution / Company*', 'Department*', 'Postal Code*', 'Country*' (a dropdown menu), and 'State / Province*' (a dropdown menu). There is a checkbox for 'I agree to the Privacy Policy'. At the bottom are 'Back' and 'Complete' buttons.

- d. Your IDA account is now created. Click IDA Home to navigate to the IDA homepage.

The screenshot shows the 'Create IDA Account' form at the 'Create Account' step, displaying a 'Welcome to the IDA' message. The progress bar at the top indicates 'Enter Email', 'Verify Email', and 'Create Account' are all complete. The message states: 'You have successfully created an IDA account. Please click the IDA Home button below to log in.' Below the message is a section titled 'Your Account Information' which displays the user's details: Name (First Name Last Name), Email (user@emailaddress.com), Institution / Company (USC LONI), Department (LONI), Postal Code (90033), Country (United States of America (The)), and State / Province (California). At the bottom right is an 'IDA Home' button.

4. DOWNLOAD DATA

Once you receive access to the study, depending upon the study, there will be up to three options under the DOWNLOAD menu: Study Data, Image Collections, Genetic Data.

1. Study Data is typically comprised of clinical assessments and analysis results data and documentation (when available from the study).
2. Image Collections contains images (MR/PET/CT, etc.) along with EEG (when available from the study).

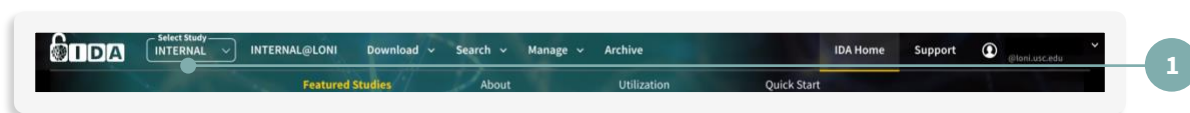
NOTE: Images must be added to a Data Collection to download. See the SEARCH section for instructions on creating Image Collections.

3. Genetic Data will contain genetic data files, often in VCF, idat or text format along with supporting documents (when available from the study).

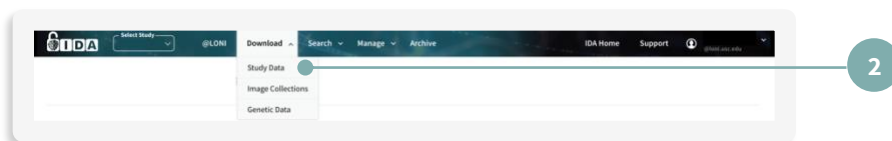
4.1 Study Data

Study data can be downloaded directly from the Study Data section in the IDA.

1. Log in to the IDA and select the study of interest from the **Select Study (1)** menu.

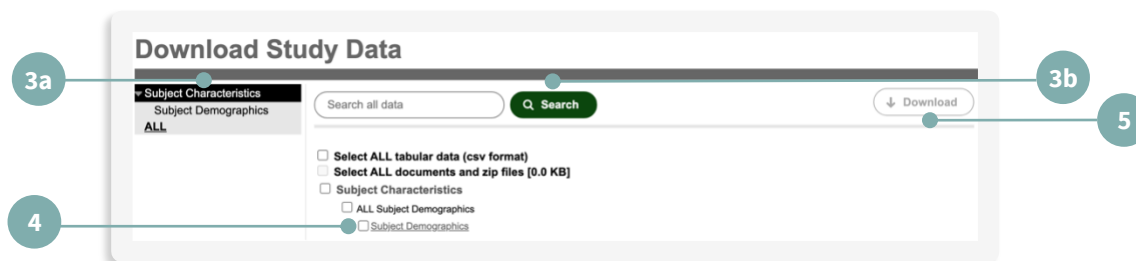


2. To download study data, clinical data, or additional study documentation, click **Study Data (2)** from the DOWNLOAD menu.



- Click the left-hand panel (3a) of the Download Study Data page to view available datasets or type in key words in the search box and click **Search** (3b). Click the dataset check box (4) and then click **Download** (5).

NOTE: You can also download the dataset by clicking the dataset title.



4.2 Image Collections

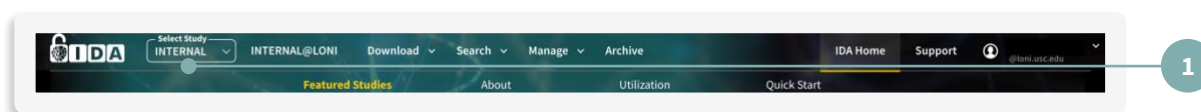
The overall process for downloading items from Image Collection is detailed in the table below:



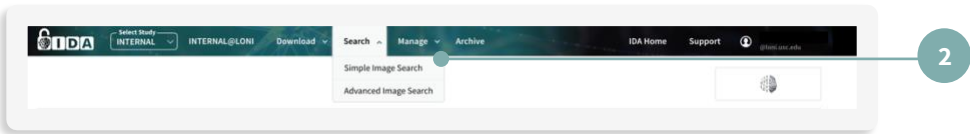
4.2.1 Simple Image Search

Simple Image Search performs a basic database query on subjects and raw images, providing information such as subject age, sex, image modality, or series description. Simple Image Search returns only original, unprocessed images. Processed images are searchable only through Advanced Image Search and may not always be available, depending on the study.

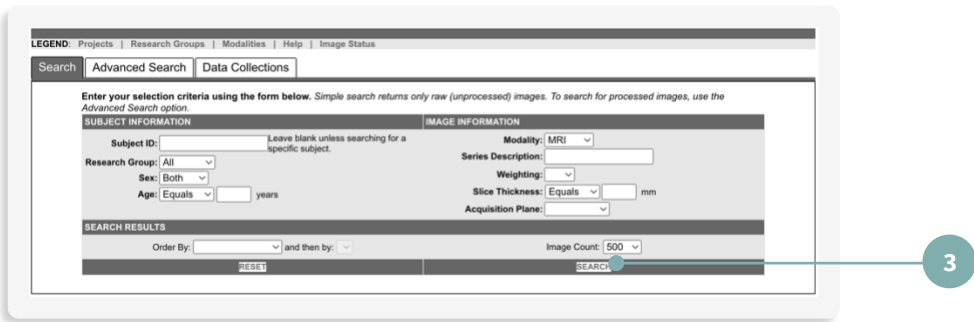
- Log in to the IDA and select a study from the **Select Study** (1) menu.



2. From the SEARCH menu, click **Simple Image Search (2)**.



3. Enter the search criteria and click **SEARCH (3)**.



4. From the Search Results page you may:
- a. Click **VIEW** to visualize scans (4).
- NOTE: This function is not available for all modalities.*
- b. Click one or more checkboxes to select items you would like download. Once selected, click **ADD TO COLLECTION (5)**.

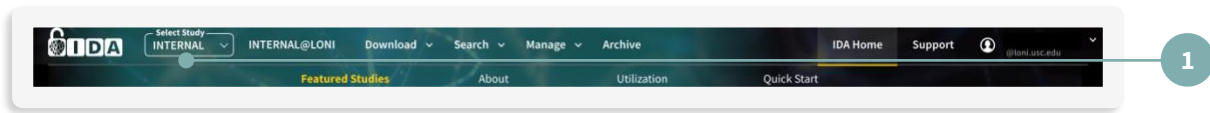


4.2.2 Advanced Image Search

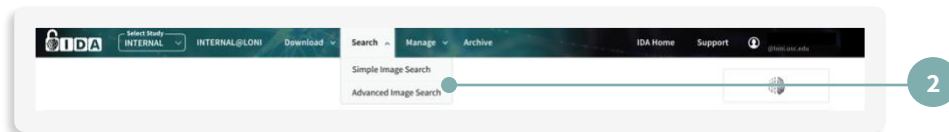
Advanced Image Search provides more extensive criteria, including processed images, and provides additional information about subjects and images. The Advanced Image Search contains Search Options in the left panel, which toggle the display of additional search criteria sections.

Steps to use Advanced Image Search:

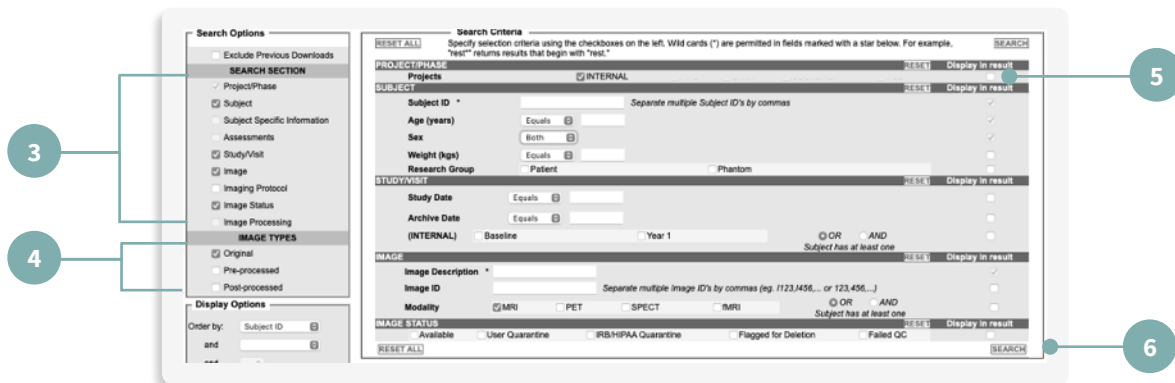
1. Log in to the IDA and select the study of interest from the **Select Study (1)** menu.



2. From the SEARCH menu, click **Advanced Image Search (2)**.



3. Use the Search Options box on the left to expand Search Criteria categories on the right.

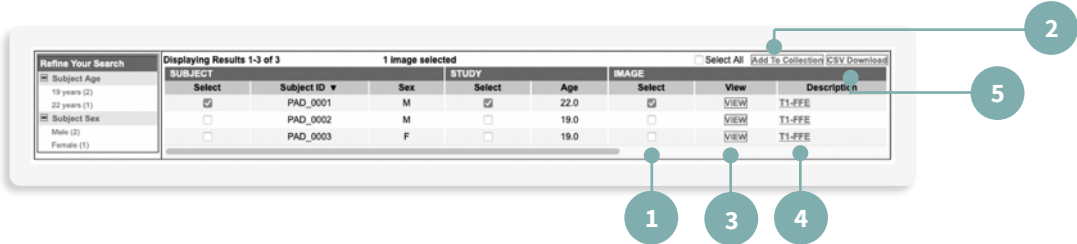


- a. Specify the categories in the **SEARCH SECTION (3)**.
- b. Specify **IMAGE TYPES (4)**. Preprocessed may be in a variety of formats.
- c. Select image search criteria using the checkboxes and drop-down menus.
- d. Select data elements to include in the results using the Display in result checkboxes (5).

NOTE: Data with corresponding grayed out boxes are displayed by default.

- e. Click **SEARCH (6)**.

The Advanced Search Results tab displays images matching your search criteria where you may:



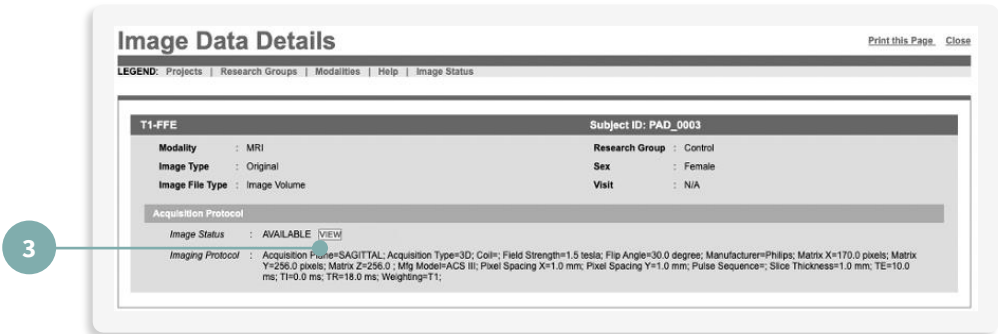
1. Click one or more checkboxes (1) to select items you would like to download. Once selected, click **Add to Collection** (2).

2. Click **VIEW** (3) to visualize scans.

3. Click the underlined **Series Description** (4) to open the Image Data Details page.

NOTE: The Image Data Details page provides additional protocol information about the image such as weighting, pulse sequence, acquisition type, etc. Processed images display an additional section describing information such as package name, process name, program name, etc. Users can also visualize images from this page.

4. Click **CSV Download** (5) to download a list of all the metadata in the Search Results list.



NOTE: You can edit the displayed metadata (or columns) by returning to Advanced Image Search and clicking the Display in Result checkbox for desired data.

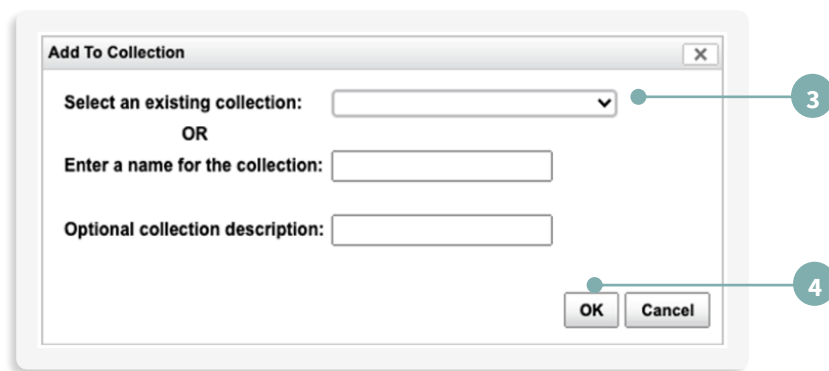
4.2.3 Create a Data Collection

Images must first be added to a Data Collection in order to be downloaded. Your personal and shared collections are located in the Collections section of the Data Collections tab.

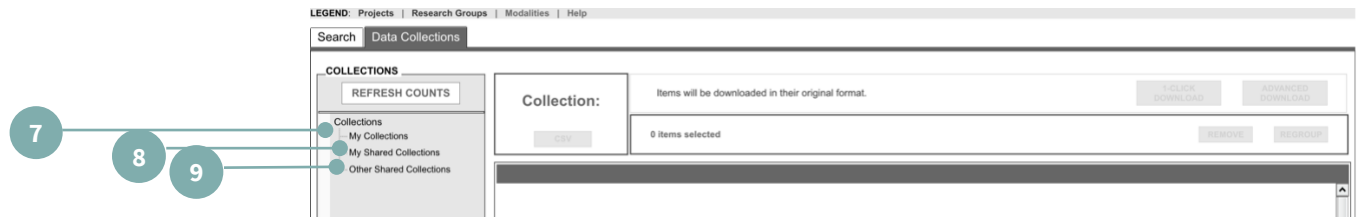
1. Perform a Simple Image Search or Advanced Image Search.
2. On the Search Results page, choose images by selecting their corresponding checkboxes (1) and click **Add to Collection** (2).



3. Enter a collection name or select an existing collection name (3) and click **OK** (4).



5. The Data Collections tab displays the collection contents. Similar to the Search Results page, the image descriptions link to the Image Data Detail page.

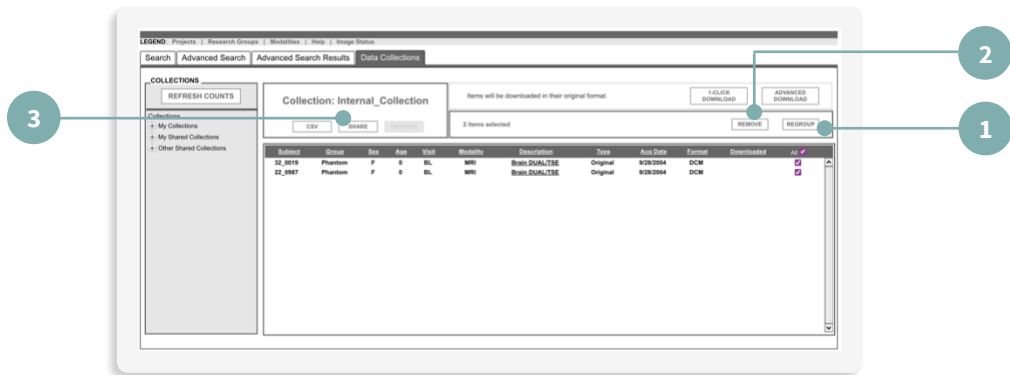


5. You can access collections in the Collections box by clicking:
 - a. My Collections - contains collections you created (7).
 - b. My Shared Collections – contains collections you have created and shared with others (Leader access level required) (8).
 - c. Other Shared Collections – contains standardized collections created by Study Leaders that are available to all users of the study (9).

4.2.4 Edit and Share a Data Collection

Data collections can be created by users of any access level, but only the owner of the collection can edit it. **Project Leader access is required to share a data collection.**

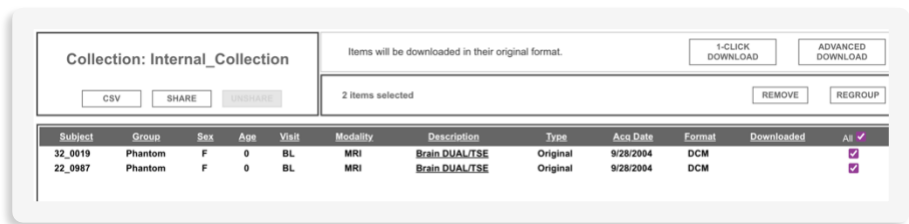
Images can be selected (using checkboxes) and moved to other collections using the **REGROUP** (1) button or deleted using the **REMOVE** (2) button. Share collections with Co-Investigators (based on user access level) using the **SHARE** (3) button. User access level can be reviewed in the ‘My Account’ section. Once a collection has been shared, it may be ‘unshared’ with the ‘UNSHARE’ button (greyed out in the screenshot).



4.2.5 Download Image Collections

Image collection data can be downloaded with the 1-Click Download or Advanced download options. Select the checkboxes of the images you want to download or use the ‘All’ checkbox to select all images within a collection.

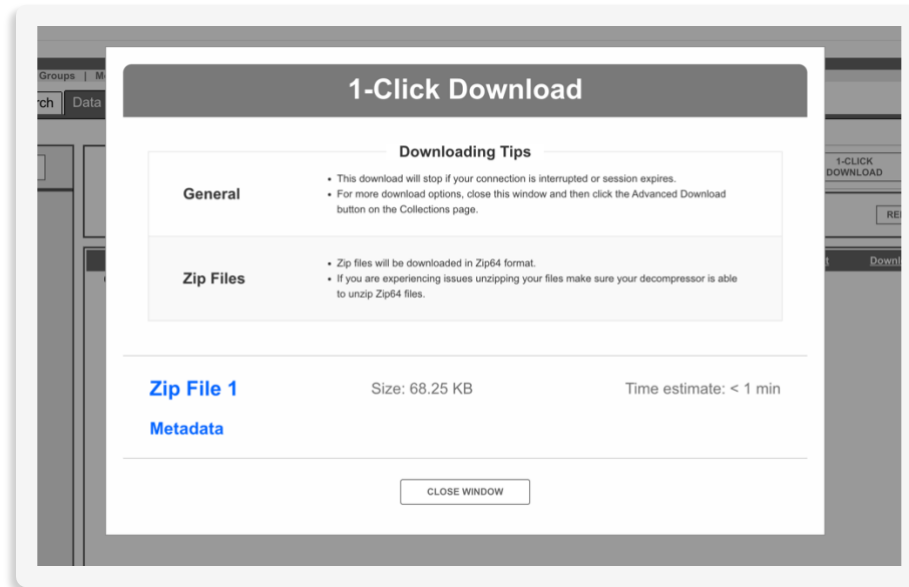
- a. Select 1-Click Download or Advanced Download (see specific instructions below).
- b. To generate a list of all the data in a Data Collection, click **CSV**.



4.2.5.1 1-CLICK DOWNLOAD

Use the built-in functionality of your web browser to quickly download all the files in your collection to a single zip file. After selecting **1-Click Download**, a pop-up window with information about your download (size and time estimate) will appear. Click on the

Download Zip File link to start downloading. The **Download Metadata** link is also available, which provides image identification details.



4.2.5.2 ADVANCED DOWNLOAD

The Advanced Download option will allow you to split your download into multiple files. Click **Advanced Download** and select a grouping option from the drop-down menu. You may click each link to download the zip files or download the list of URLs to use with the IDA Downloader for error-checking and automatic retries.



4.2.5.3 IDA Downloader

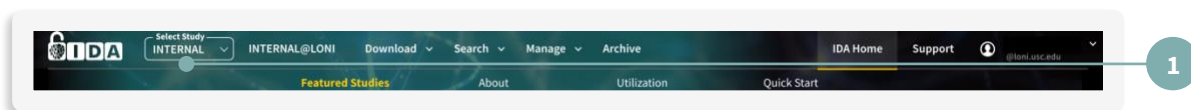
The IDA Downloader is a Java jar file that incorporates error-checking and automatic retries into a command line executable. When local download speeds and connectivity are problematic, we recommend downloading data sets in smaller batches and using the IDA Downloader.

To use the IDA Downloader, Java version 12 or higher must be installed on your computer. The IDA Downloader can be downloaded [here](#). Please refer to the README.md file included in the IDA Downloader zip file for the command line syntax for executing the IDA Downloader jar file.

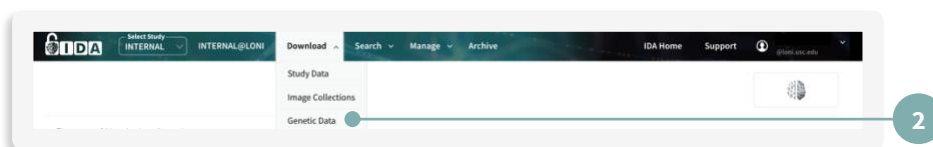
4.3 Genetic Data

Genetic data can be downloaded directly from the Genetic Data section in the IDA.

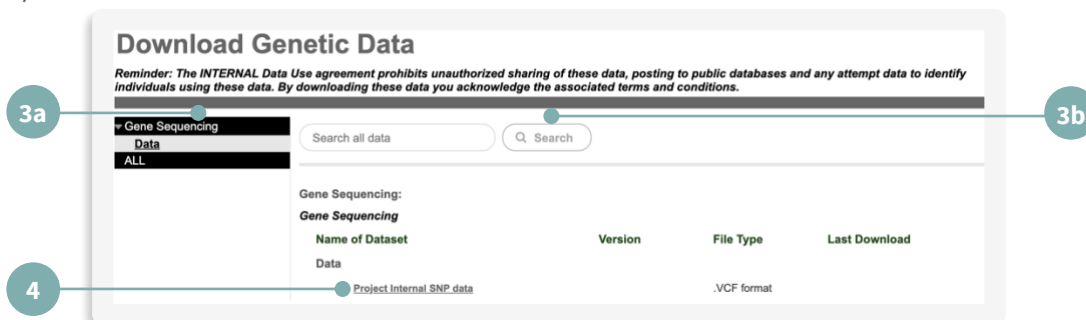
1. Log in to the IDA and select the study of interest from the **Select Study** (1) menu.



2. To download genetic data, click **Genetic Data** from the DOWNLOAD menu.



3. Click the left-hand panel (3a) of the Download Genetic Data page to view available datasets or type in key words in the search box and click **Search** (3b). Click the dataset title (4) to download.



5. UPLOAD (ARCHIVE) DATA

There are two steps in the archive process: de-identification and file transmission. The de-identification step removes or replaces potentially identifying subject information from the image headers. Once the results of the de-identification process have been validated, the files are transmitted from the user’s local computer to LONI. Upon arrival at LONI, the data are stored in a fault-tolerant storage area network and the database is populated with relevant metadata attributes.

There are two options for uploading images – Single and Batch Archive, each with its own distinct advantages.

Batch Archive	VS	Single Archive
Archive image files from multiple subjects simultaneously.		Archive image files from only one subject at a time.
Less time-consuming alternative for archiving multiple studies.		Allows for more user interaction.
Automated Process requiring minimal interaction.		Examine list of de-identified files and corresponding header information before transmission.
Subject and phantom scans acquired on a single day may be uploaded in the same session.		Select which series should be archived when multiple series are available.

REQUIREMENTS:

- A computer with internet access (standard internet ports 80 and 443) and ~136 MB disk space
- Computer rights to install new applications
- IDA-Uploader 2.0 application

NOTE: Before any processed images can be uploaded, the original raw images must be archived. Only select users are given permissions to upload processed images. Contact the IDA team for more information.

5.1 OBTAINING AND INSTALLING THE IDA-UPLOADER

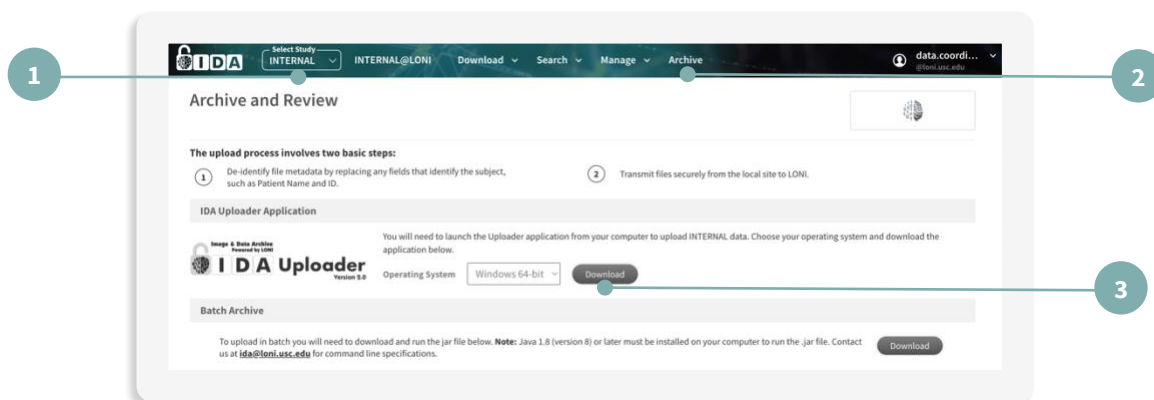
5.1.1 [Installing the IDA-Uploader for Windows](#)

5.1.2 [Installing the IDA-Uploader for Mac](#)

5.1.3 [Installing the IDA-Uploader for Linux](#)

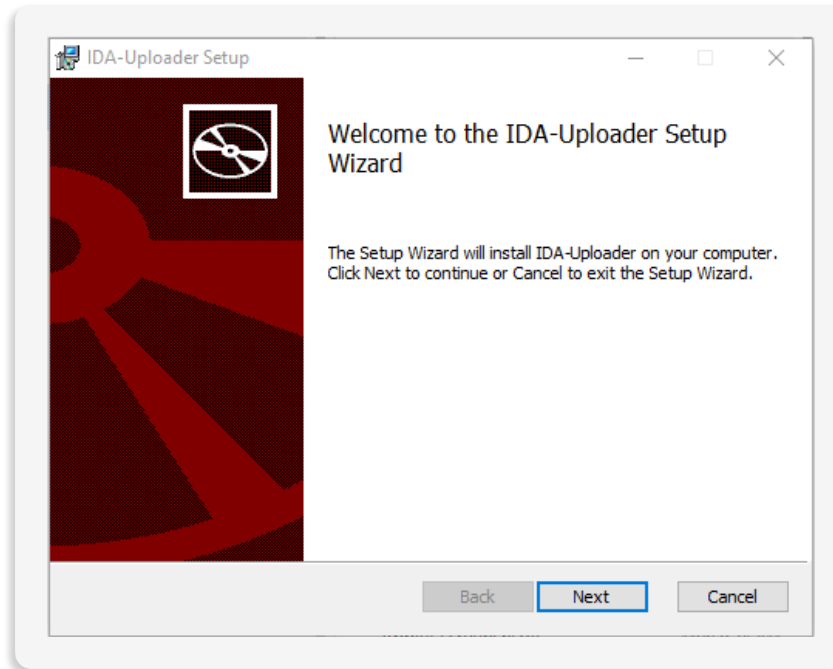
5.1.1 Installing the IDA-Uploader for Windows

1. Log in to the IDA and select the study of interest from the **Select Study (1)** menu.
2. Select the **Archive (2)** menu option.
3. Select your operating system (Windows 64-bit) from the dropdown menu.
4. Click **Download (3)**.

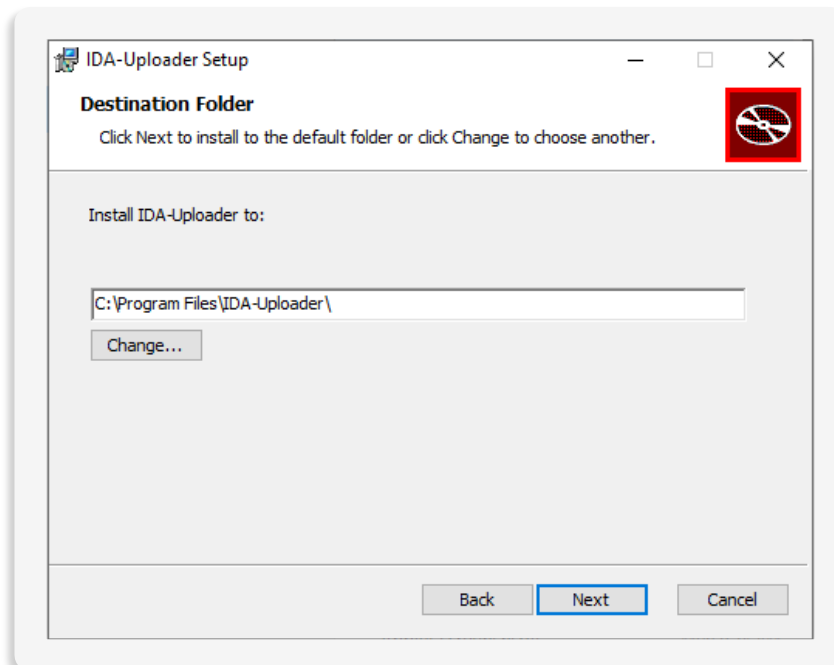


5. Open the application by clicking on the download in your browser or by locating the IDA-Uploader-2.0.msi application in the Downloads section of your File Explorer.

6. You will be taken to the IDA-Uploader Setup Wizard – click **Next** to continue.

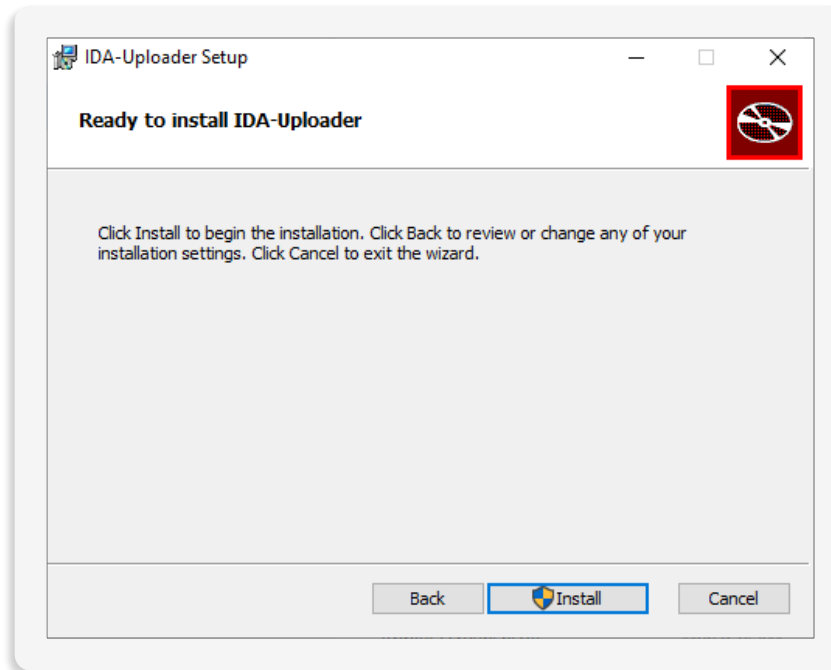


7. Choose your destination folder and click **Next**.

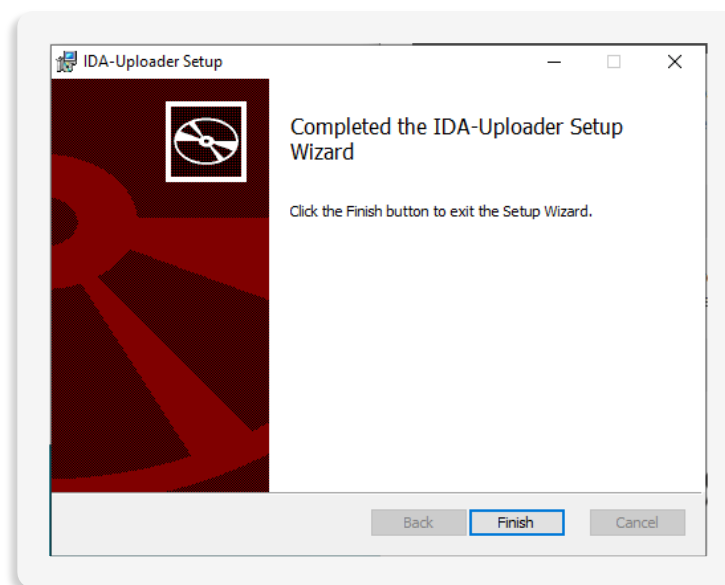


8. Click **Install**.

*NOTE: If you receive a popup asking for permission to run the application, click **Yes**.*



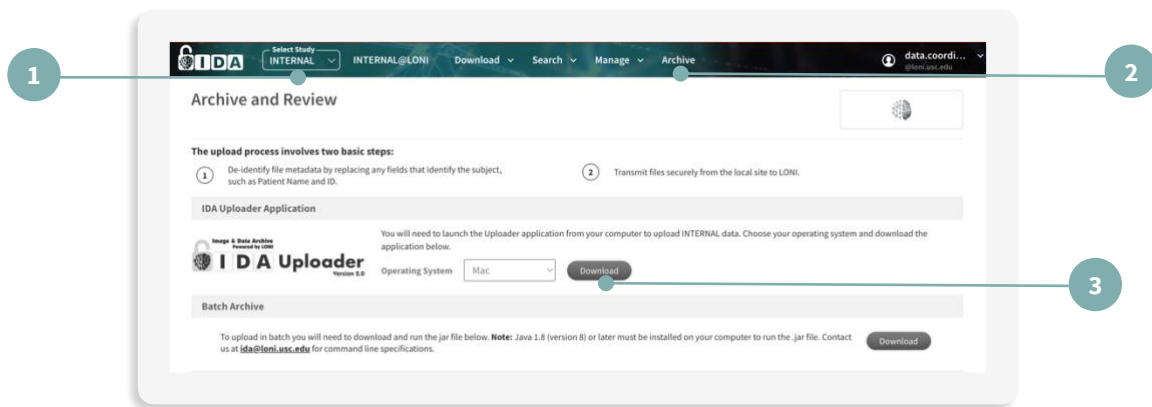
9. Click **Finish** to exit the Setup Wizard.



10. Locate the application on your device by typing in **IDA-Uploader** in the Windows Start Menu.

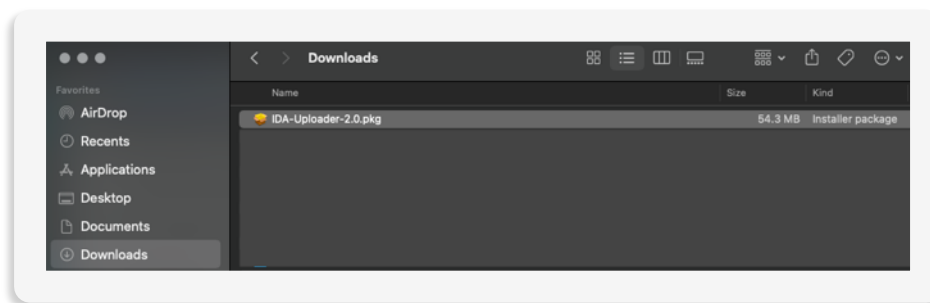
5.1.2 Installing IDA-Uploader for Mac

1. Log in to the IDA and select the study of interest from the **Select Study (1)** menu.
2. Click **Archive (2)** from the menu.
3. Select your operating system (Mac) from the dropdown menu.
4. Click **Download (3)**.

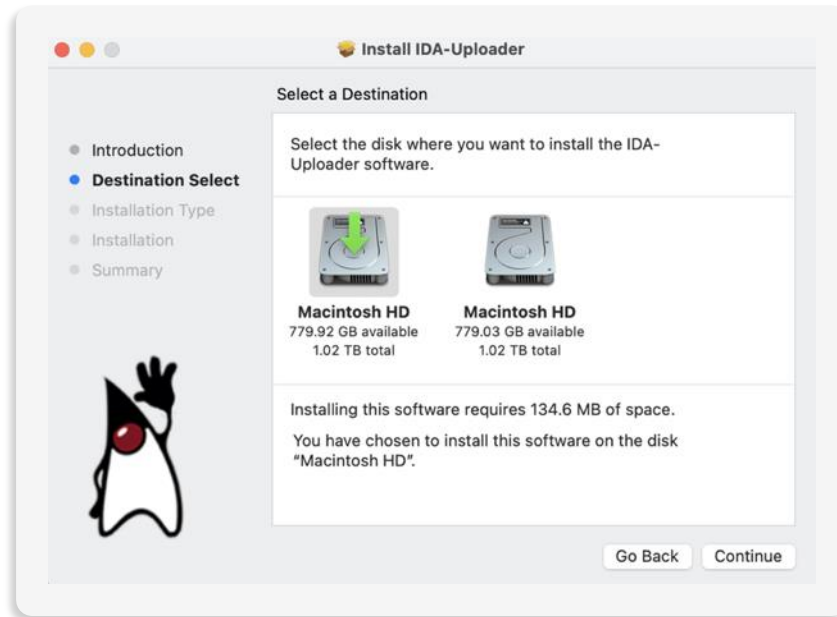


5. Open your Finder application, click Downloads, right-click **IDA-Uploader-2.0.pkg** and select **Open**.

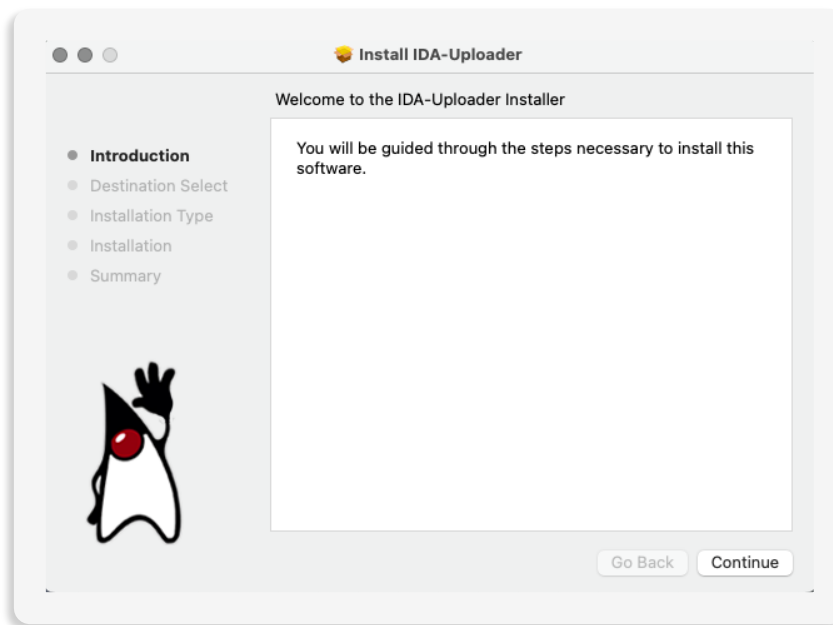
NOTE: If you try to install by double-clicking on 'IDA-Uploader-2.0.pkg', you may receive a message that it cannot be opened. Please ensure that you follow the instructions above to successfully open.



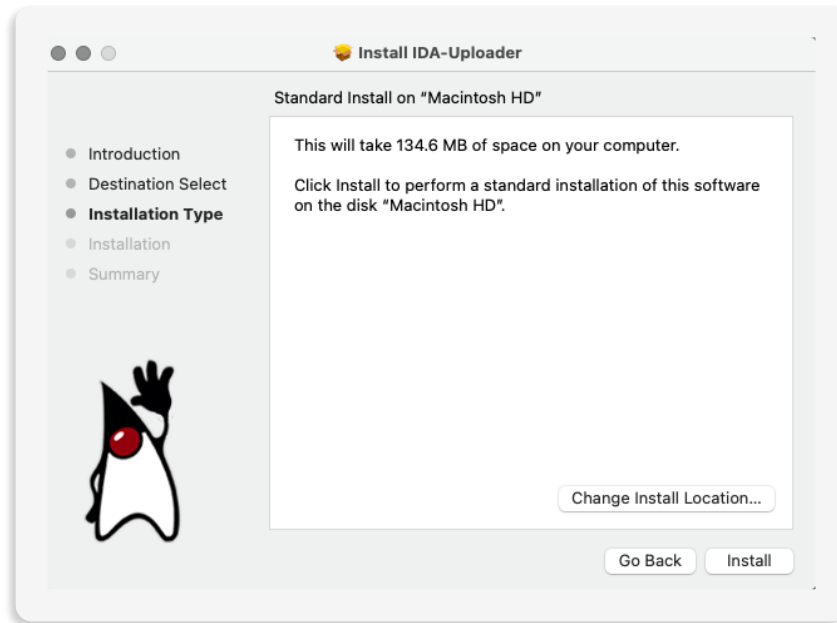
6. An Install IDA-Uploader window will appear. Click **Continue**.



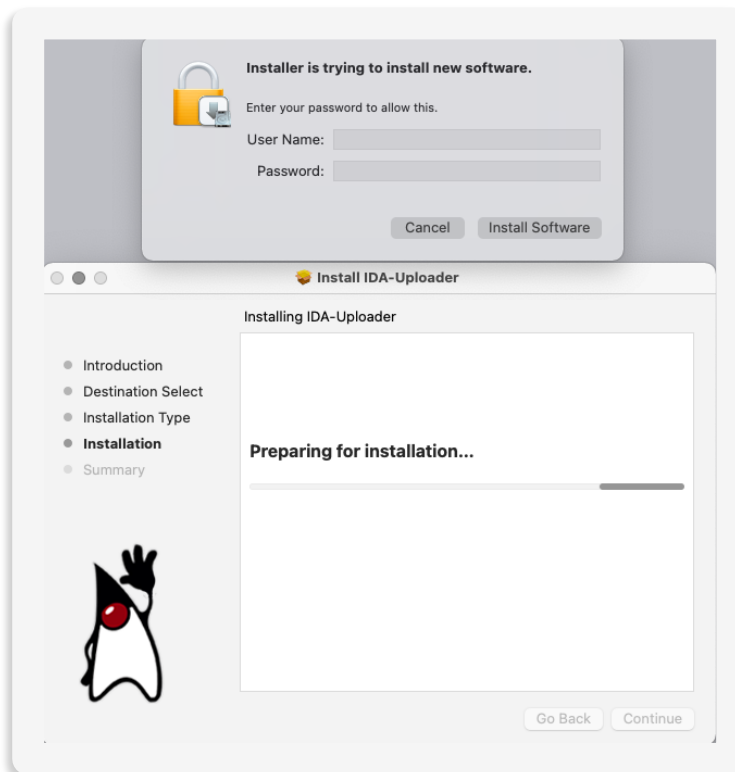
7. Next, choose a destination for the installation. Click **Continue**.



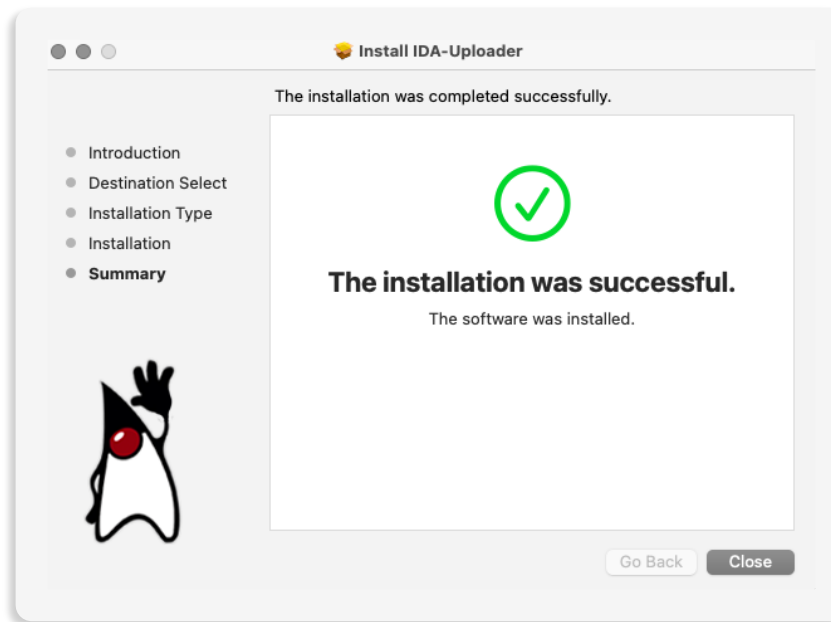
8. For Installation Type, you can review the details of the installation. Click **Install**.



9. During the installation, you will need to enter the User Name and Password for the computer. Click **Install Software**.



10. Once the installation is complete, the window will provide a summary of a successful installation. Click **Close**.



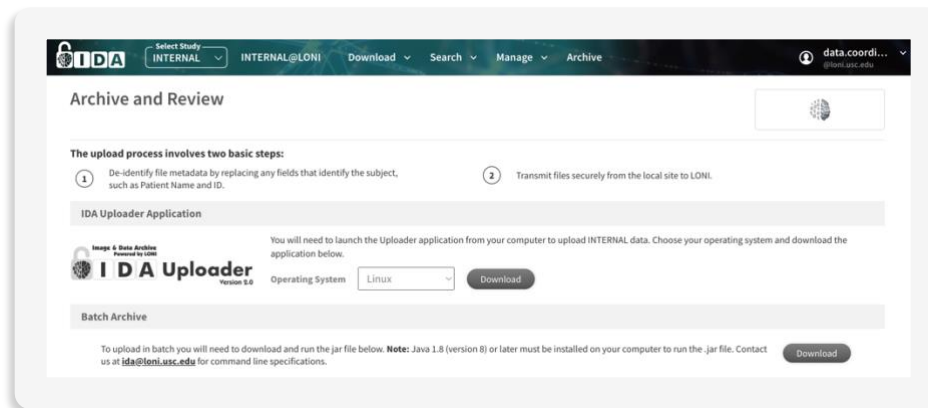
NOTE: A window will appear to ask, “Do you want to move the “IDA-Uploader” installer to the trash?”. You can choose Keep or Move to Trash.

11. Locate the application in your computer by opening your Finder application. Click on Applications and double-click IDA-Uploader.

5.1.3 Installing IDA-Uploader for Linux

1. Log in to the IDA and select the study of interest from the **Select Study (1)** menu.
2. Select the **Archive (2)** menu option.
3. Select your operating system (Linux) from the dropdown menu.
4. Click **Download (3)**.
6. Please visit <https://www.oracle.com/java/technologies/javase-downloads.html> to download the latest Oracle JDK.

NOTE: A minimum version of 15.0.1 is required to run IDA-Uploader-2.0.jar

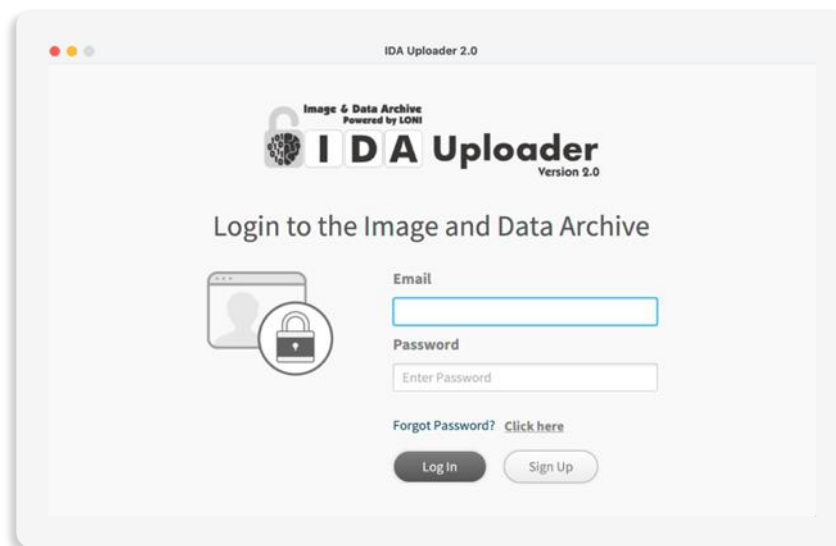


6. Choose DEB, PRM or the compressed archive depending on your Linux Distribution.
7. Once you have completed the installation, open the terminal and run `java -jar IDA-Uploader-2.0.jar`

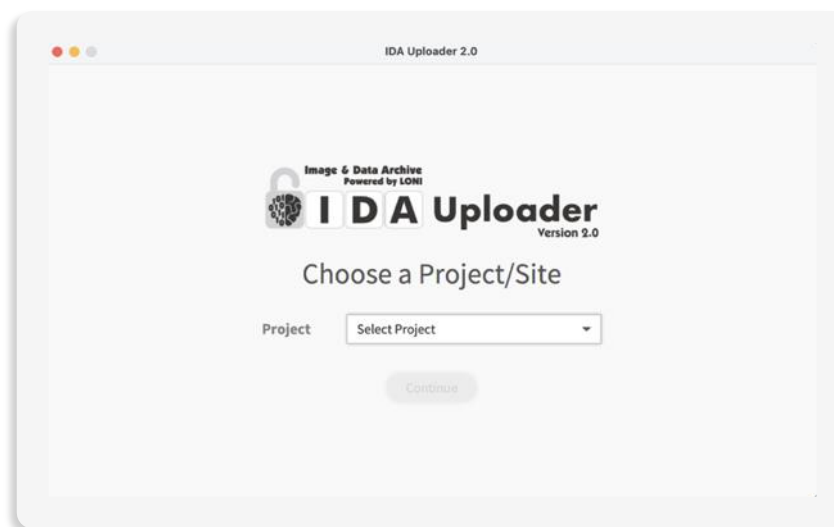
5.2 Single Archive

The single archive option can be used when uploading studies for the same participant.

1. Open the IDA-Uploader application.
2. Enter your email and password, then click **Log In**.



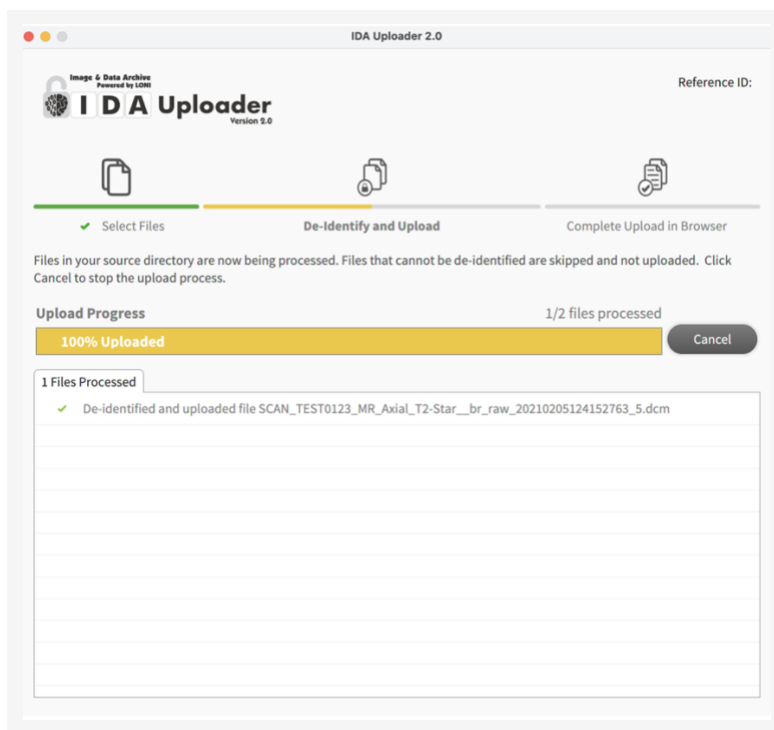
3. Select your study from the dropdown menu. Once you select your study, a dropdown with the sites will appear. Select your site and click **Continue**.



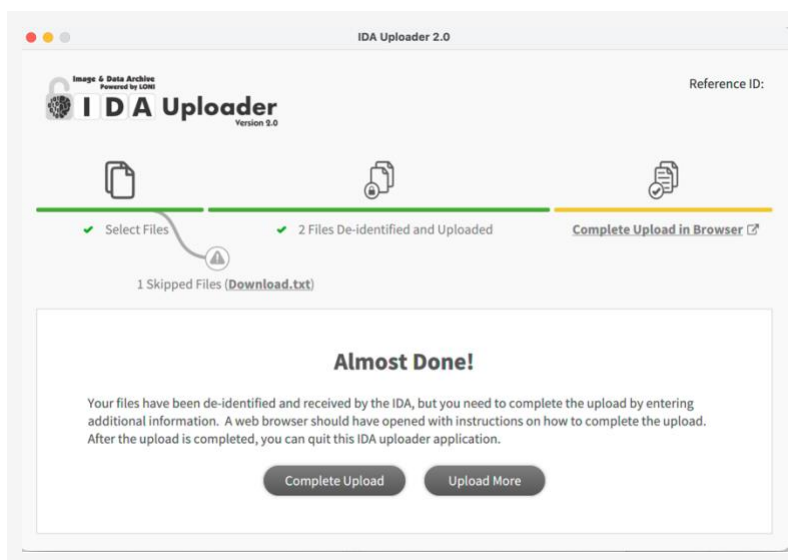
4. Enter the Subject ID. Click **Browse** to select the Source Directory. Then click **Upload**.
NOTE: The Source Directory is the directory containing the files to be uploaded. If your Source Directory contains subdirectories, choose to include/exclude those files by checking “Search subdirectories”.

A screenshot of the IDA Uploader 2.0 application window, showing the 'De-identify and Upload' step. The window has a title bar and a progress bar at the top with three stages: 'Select Files' (highlighted), 'De-identify and Upload', and 'Complete Upload in Browser'. The main content area is titled 'Enter required information below and click Upload to begin the process.' It contains three sections: 'Site' with the text 'University of Southern California'; 'Subject ID' with a description 'De-identifier to replace patient ID. Maximum of 15 characters allowed.' and an input field labeled 'Enter Subject ID'; and 'Source Directory' with a description 'Location of files to de-identify and upload.', a 'Click Browse to select directory' button, and a 'Browse' button. There is also a checkbox for 'Search subdirectories'. At the bottom, there is a link 'Need help? Email ida@loni.usc.edu', a 'Cancel' button, and an 'Upload' button.

7. You will be able to see the progress of your upload in the De-identify and Upload section.



8. Once the files are de-identified and transferred to the IDA, you will need to complete the upload in your web browser.

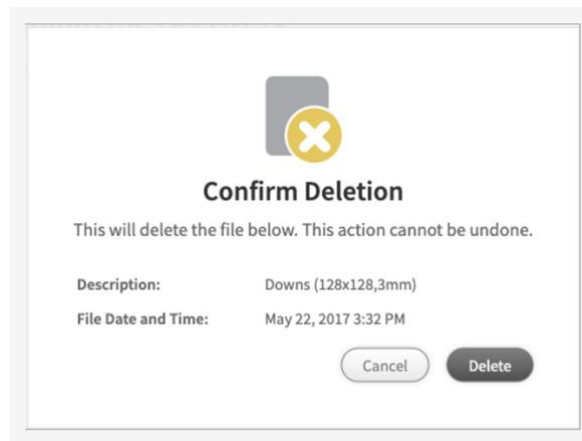


9. Your web browser should automatically open a new "Login and Continue" page. Enter your IDA email and password. Click **Log In**.

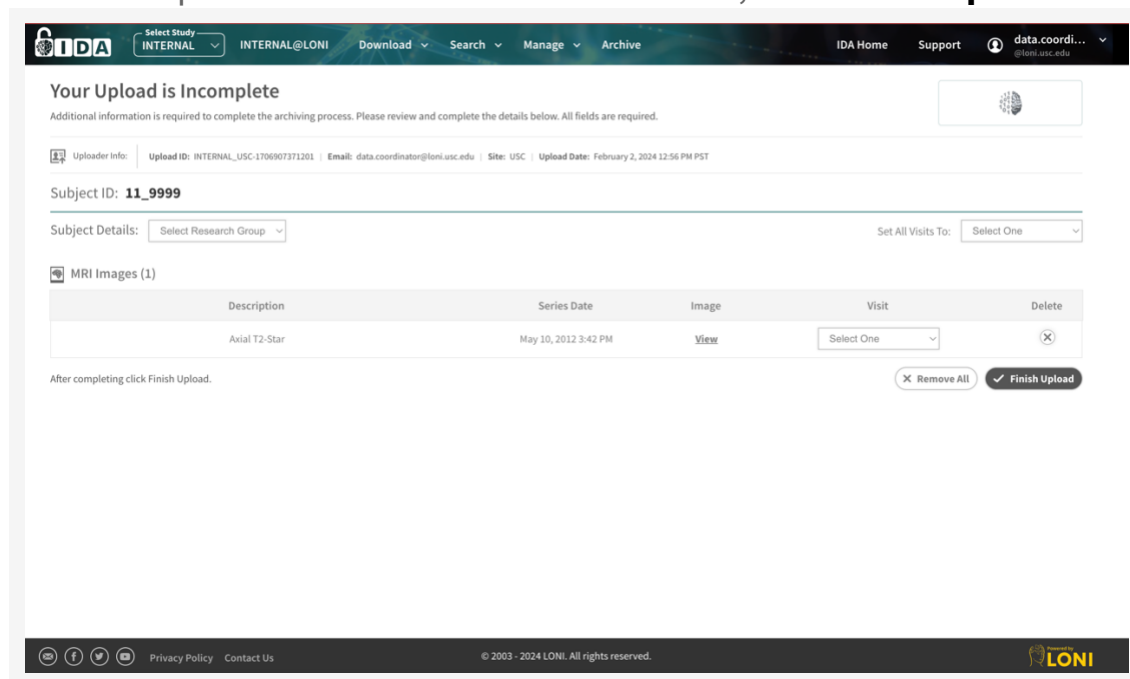
*NOTE: If your web browser does not automatically open the “Login and Continue” page, please click **Complete Upload** and you can copy-paste the link in your browser.*

10. In the “Your Upload is Incomplete” page, you can review the details of your upload.

*NOTE: If any data was mistakenly uploaded, you can click the **X** in the Delete column and it will be deleted. A window will appear to confirm the deletion. Click **Delete** to confirm file deletion or **cancel**.*

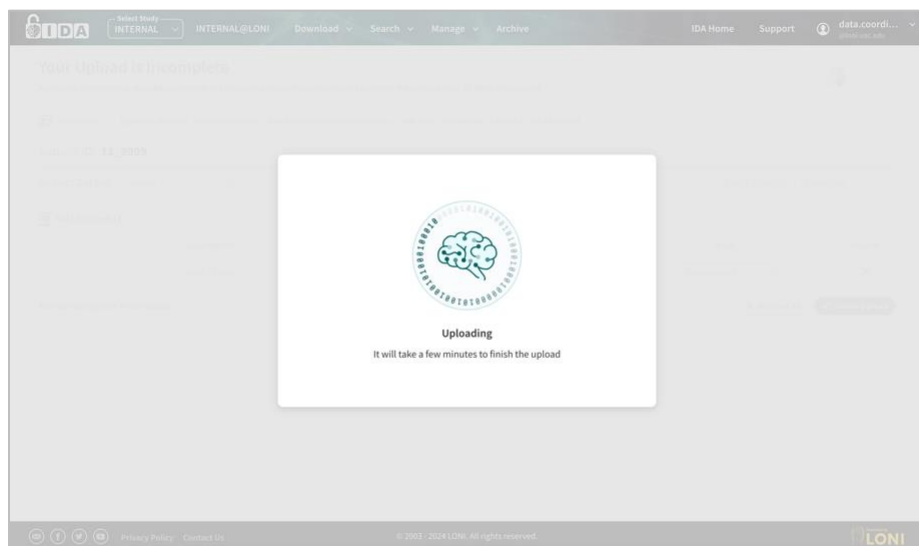


11. Once all required information has been entered, click **Finish Upload**.

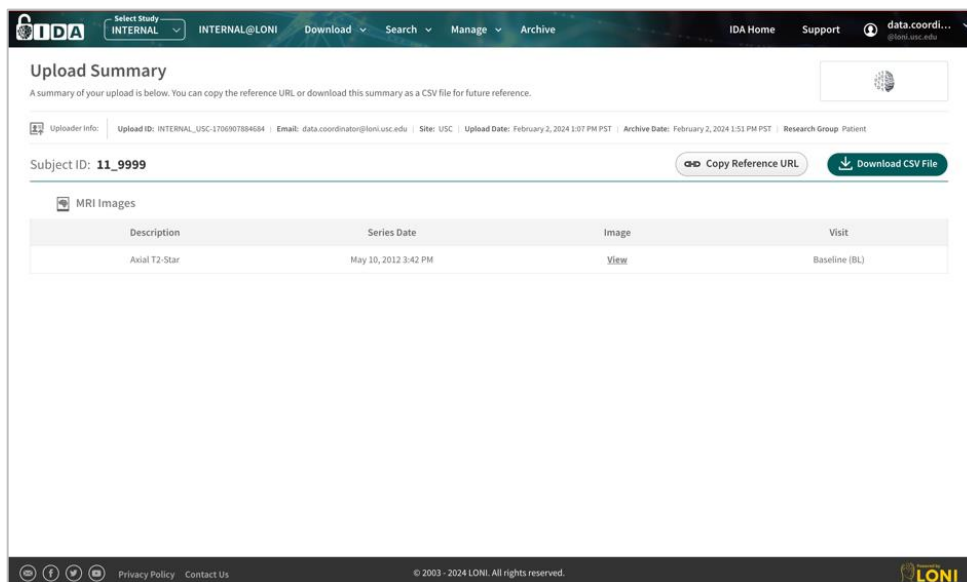
A screenshot of the "Your Upload is Incomplete" page in the LONI IDA system. The page has a dark header with the LONI IDA logo and navigation links. The main content area shows upload details for "Subject ID: 11_9999". It includes a table for "MRI Images (1)" with columns for Description, Series Date, Image, Visit, and Delete. The table contains one row for "Axial T2-Star" with a "View" link and a "Delete" button. Below the table are "Remove All" and "Finish Upload" buttons. The footer contains social media links, privacy policy, and copyright information.

Description	Series Date	Image	Visit	Delete
Axial T2-Star	May 10, 2012 3:42 PM	View	Select One	X

12. You will see a screen that informs you that the upload is being processed.



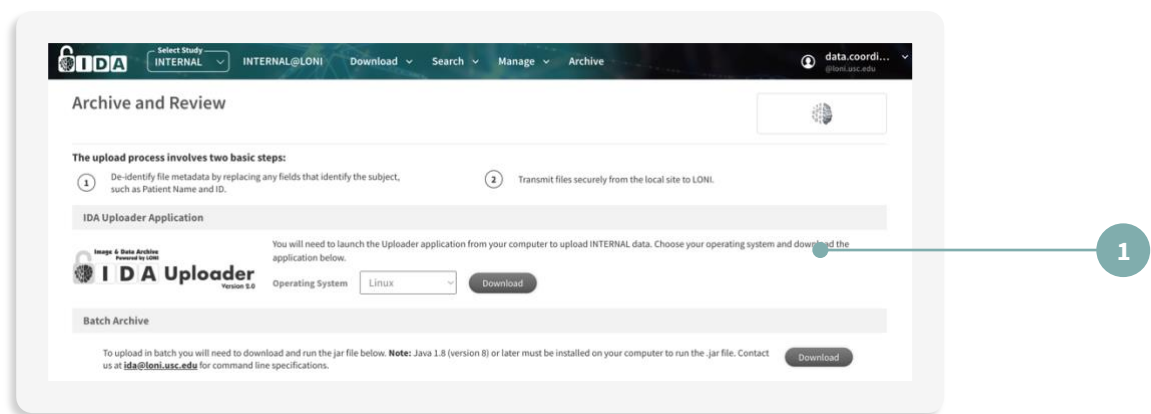
13. Once the upload is processed, a summary page for the upload will be displayed. Please note that you have the option to click **Download CSV Files** to keep a record of the upload.



14. You can close the IDA-Uploader application or to upload images for another subject, click **Upload More** in the IDA-Uploader application.

5.3 Batch Archive

Although the Batch Archive process is convenient for uploads of studies from multiple participants, it is recommended for users skilled at operating from the command line/terminal. The Batch Archive process is similar to Single Archive, except that multiple subjects and image series can be submitted in a batch. Batches can be of the same or different file formats and modalities. However, users cannot review the results of the de-identification process prior to the batch upload.



REQUIREMENTS:

- IDA Batch Uploader JAR file, available for download (1) from the IDA Archive page
- Command line instructions
- Template file

Email ida@loni.usc.edu for batch archive instructions and template file.

6. TROUBLESHOOTING

Please refer to the [IDA Support](#) page for troubleshooting assistance.

7. GLOSSARY

Term	Definition
Advanced Image Search	Allows users to search with a combination of subject, study, and image descriptors (including processed images). Users can search several modalities simultaneously and save search results.
Batch Archive	Allows data to be uploaded from multiple subjects simultaneously; Batches may be different file formats and modalities
Collections	Data selected from the results of a Simple or Advanced Image Search and given a meaningful description as the collection name
De-identification	Removal or replacement of certain image header contents to protect the identity of the subject
DICOM	Digital Imaging and Communication in Medicine
DTI	Diffusion Tension Imaging
Download	The ability to transfer data collections via 1-CLICK DOWNLOAD and ADVANCED DOWNLOAD options
Edit Metadata	Modifying information such as sex, age, and weight, as well as other project-specific attributes
FMRI	Functional Magnetic Resonance Imaging

Image metadata	Information contained in the image header describing the subject and image
MRA	Magnetic Resonance Angiography
MRI	Magnetic Resonance Imaging
Original data	Raw images files pre- or post-processed
PET	Positron Emission Tomography
Processed data	Image data files transformed or modified from their original form, i.e, registered, warped or corrected data
Series description	An identifier (name) given to a data set
Simple Image Search	Searches based on a limited set of criteria. Returns only unprocessed images.
Single archive	Upload of data for one subject at a time; Files of the same subject can be uploaded together, regardless of format.
Source directory	Directory containing the files to be uploaded.
Target directory	The Target Directory can be an existing or a new directory; A temporary location to contain de- identifies files, information and a copy of the logs of the upload process
Upload (Archive)	The process of securely transmitting de-identified data to be stored in the LONI Image & Data Archive for future retrieval